

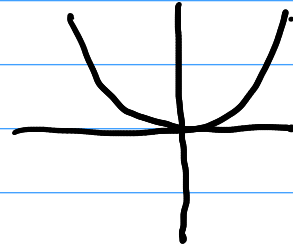
$m=2$
 $\leftarrow P(x) = \cancel{x^2} + 2\cancel{x} + 1$
 $\leftarrow$ degree-5 poly
 $P(x) \cdot Q(x) = *x^5 + *(x^4) + *x^3$

$\leftarrow (x^3) + (x^2) + x + 1 = Q(x)$
 $n=3$
 $a_1 b_3$

$(1 \times 1 + 1 \times 2) x^4 \leftarrow 3x^4$

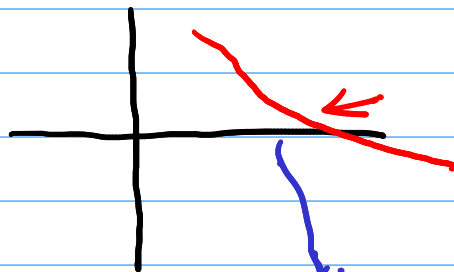
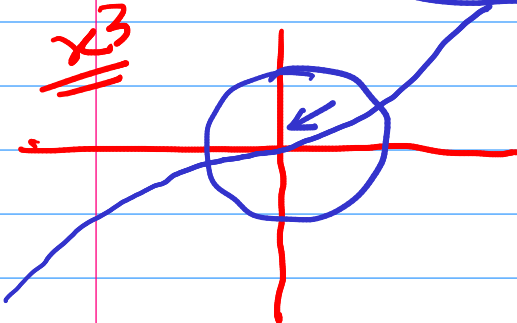
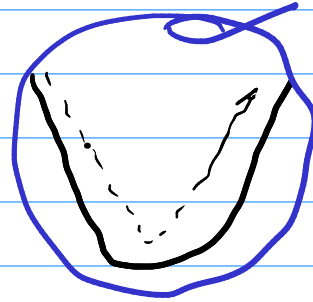
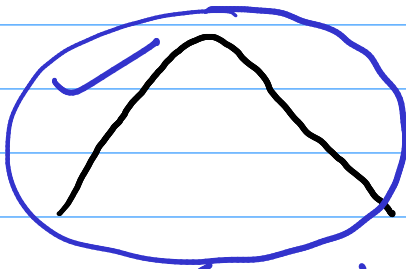
$P(x) = \underline{a_0 + a_1 x + a_2 x^2}$

$P(x) = x^2$

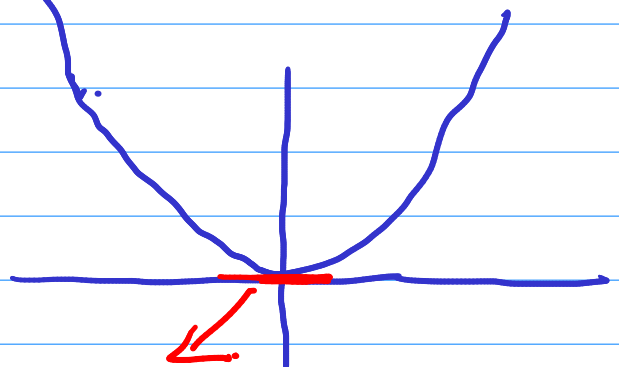


$\star P(x) = 7x^{\textcircled{1}} + 9x^5 + 1$; $x \rightarrow -\infty, P(x) \rightarrow -\infty$
 $= 7x^0 + 9x^5 + 1 = 7 + 9x^5 + 1$

$\text{If } r=0 \Rightarrow P(x) = 9x^5 + 8$

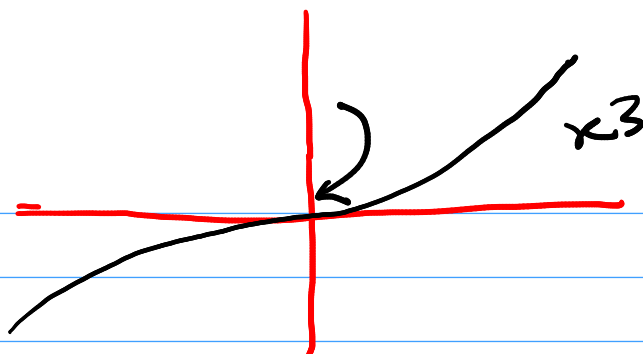
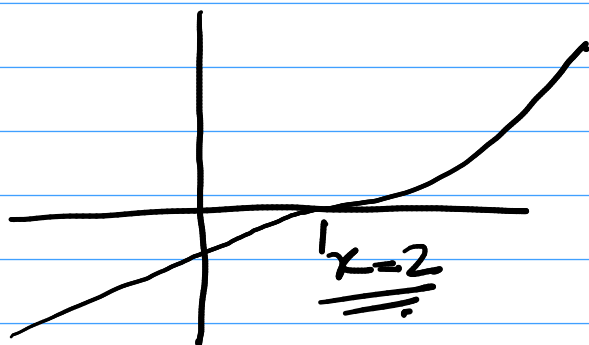


$P(x) = x^4$ - degree-4



at most $\textcircled{n-1}$
 (exactly)

A. 13-Q.1
12 Q-3



⇒ I.M.V.T. $P(x)$

$[a \ b]$
 $\uparrow$
 x

$$P(a) = m. < P(b) = n.$$

$$m. < n. < n.$$

$$P(x) = n.$$

$$P(3) = -2.$$

$$P(2) = 1.$$

$$\underline{\underline{2 < x < 3}}$$

$$-2 < 0 < 1$$

$$\underline{\underline{P(x) = 0}}$$

$$\underline{\underline{3}} = 2.999 \dots \dots \dots \checkmark \alpha + \beta + \gamma = *$$

$$2. \underline{998} \underline{9998} \underline{99998} \checkmark \alpha \beta \gamma = *$$

$$\checkmark \alpha \beta + \beta \gamma + \gamma \alpha = *$$

$$2x^3 - 18x^2 - 24x + 40$$

$$\curvearrowright \curvearrowright x = P/Q$$

$$P \mid 1. \leftarrow \pm 1$$

$$Q \mid 2. - \pm 1, \pm 2.$$

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Rational root theorem:

$$a_n x^n + a_{n-1} x^{n-1} + \dots + a_0 = f(x)$$

If $x = p/q$ is a root of $f(x)$.

$$\Rightarrow p | a_0 \text{ \& } q | a_n.$$

$$P(x) = (x+4)^5 (2x^2+4) = 0$$

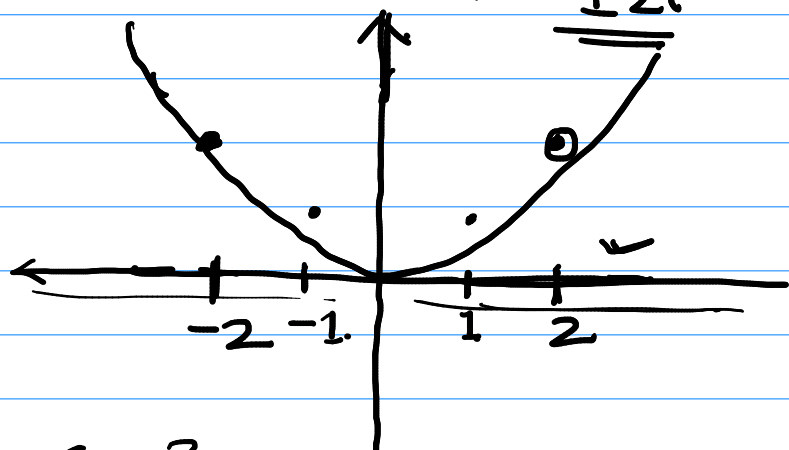
$$\Rightarrow 2x^2+4=0 \Rightarrow x^2 = -2$$

$$x = \pm 2i$$

Even function.

$$f(-x) = f(x)$$

$$f(-2) = f(2)$$



$$f(x) = x^2.$$

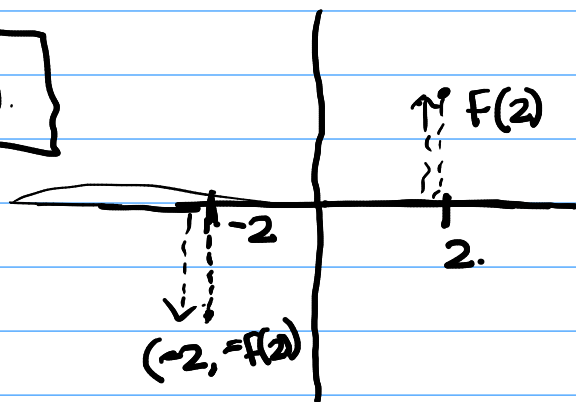
$$f(-x) = (-x)^2 = x^2 = f(x).$$

$$f(x) = x^4$$

odd:

$$f(-x) = -f(x)$$

$$f(-2) = -f(2)$$



$$f(x) = x^3.$$

$$f(-x) = (-x)^3 = -x^3 = -f(x)$$

$$f(x) = x^2 + x \leftarrow f(-x) = x^2 - x \neq f(x) = x^2 + x \neq -f(x) = -x^2 - x$$

Lect 1

$$P(x) = (x^2 + x + 1) \underline{(x+1)} \underline{x}$$

$x, x+1, x^2+x+1, \underline{x(x+1)}, x(x^2+x+1), (x+1)(x^2+x+1), P(x),$

★ $\{2\}$ $\{\underline{\underline{2}}\}$ $\{\}^0$
 $\{\underline{\underline{\{ \} }}\} \rightarrow 1$

Graph of Polynomial

⇒ Roots ⇒ multiplicities of those Roots
 ⇒ End behavior
 ⇒ Y-Intercept.

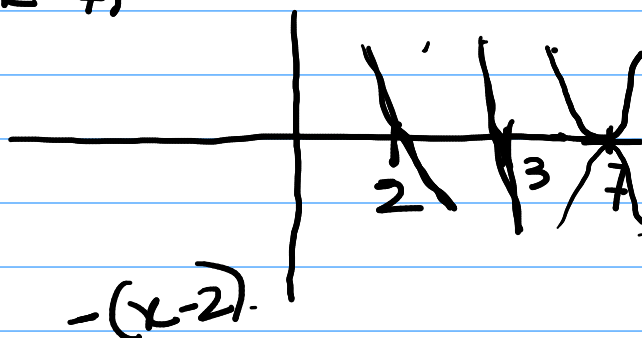
$(x-a)$

$$P(x) = (x-2)^3 (x-3)^5 (x-7)^4$$

mult. 2 → 3 - odd

mult. 3 → 5 - odd

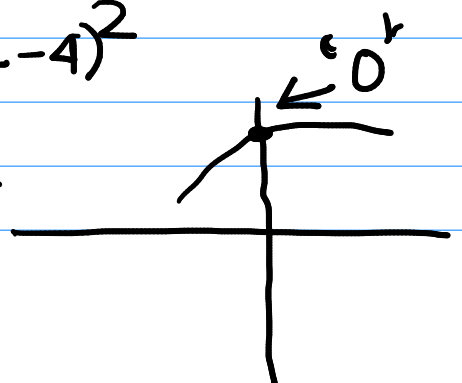
mult. 7 → 4 - even.



$$P(x) = (x-2)^3 (x-3) \underline{\underline{-(x-2)}} (x-4)^2$$

$$= -(x-2)^4 (x-3) (x-4)^2$$

$x=0$ ⇒ $P(0)$ ^{Int} $(0, P(0))$



End behavior

$$P(x) = x^5 + 2x^2 + x + 7$$

$$x^5 \rightarrow -\infty \text{ as } x \rightarrow -\infty$$

$$x^5 \rightarrow +\infty \text{ as } x \rightarrow +\infty$$



$$P(x) \rightarrow -\infty \text{ as } x \rightarrow -\infty$$

$$\rightarrow +\infty \text{ as } x \rightarrow +\infty$$

$$P(x) = -2x^4 + x^3 + 1$$

$$x^4 \rightarrow +\infty$$

$$-2x^4 \rightarrow -\infty$$

$$x \rightarrow +\infty : -2x^4 \rightarrow -\infty$$

$$x \rightarrow -\infty : -2x^4 \rightarrow -\infty$$

$$x^4 \rightarrow +\infty$$

$$-2x^4 \rightarrow -\infty$$

$$P(x) = -1(x-2)^2(x+3)^3(x+1)(x-5)$$

$$-x^7 + (\dots)$$

Roots: $-3, -1, 2, 5$

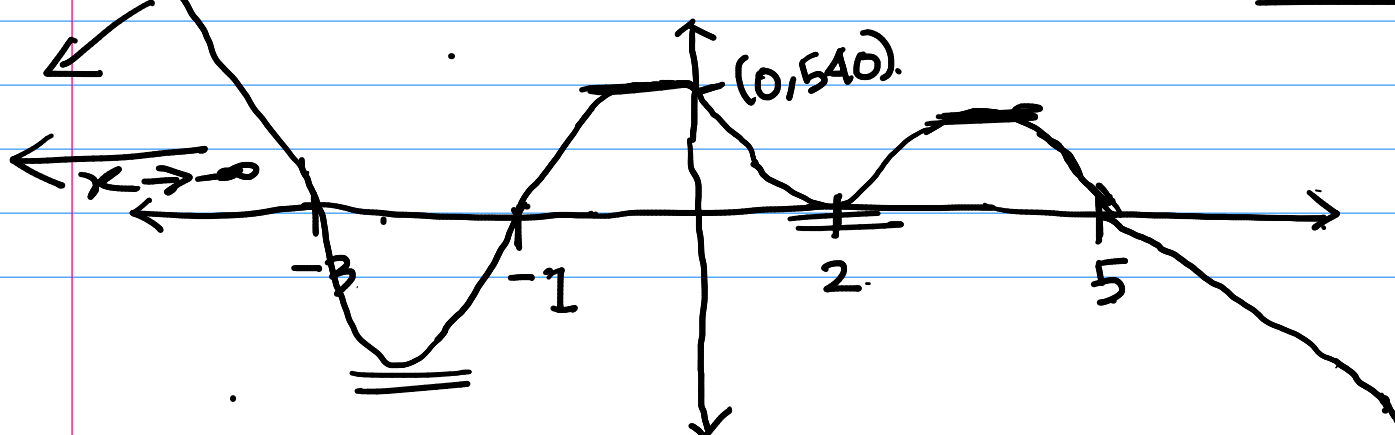
$\downarrow 3$ $\downarrow 1$ $\downarrow 2$ $\downarrow 1$

Y-Inter: $P(0) = -4 \times 27 \times -5 = 540$

$(0, 540)$

End beh: as $x \rightarrow +\infty : -x^7 \rightarrow -\infty : P(x) \rightarrow -\infty$

$P(x) \rightarrow +\infty$ as $x \rightarrow -\infty ; -x^7 \rightarrow +\infty$ $\underline{\underline{P(x) \rightarrow +\infty}}$



(a) x^*

$$x^2: \quad \begin{array}{ll} x \rightarrow +\infty & x^2 \rightarrow +\infty \\ x \rightarrow -\infty & x^2 \rightarrow +\infty \end{array}$$

$$\textcircled{-2x^2}: \quad \begin{array}{ll} x \rightarrow +\infty & -2x^2 \rightarrow -\infty \\ x \rightarrow -\infty & \rightarrow -\infty \end{array}$$

